

The Complexity–Spuriousness Trade-off in Probing: Toward a Unified Capacity Measure for Diagnostic and Causal Interpretability

Tiago Pimentel
ETH Zürich
`tiago.pimentel@inf.ethz.ch`

June 2, 2026

Abstract

Diagnostic probing and causal probing (via distributed alignment search, DAS) both tend to find what one looks for if the probe is powerful enough. [Pimentel et al. \(2020\)](#) show that a sufficiently powerful diagnostic probe always recovers a target property, and [Sutter et al. \(2025\)](#) show that a sufficiently complex alignment map makes any algorithm a distributed abstraction of any network. Both are statements about a limit: at infinite complexity, either method can fit any target, including a random one, so its reported score says nothing about the network. Read at face value, the causal result says that causal probing is no better than diagnostic probing—it leaves us where diagnostic probing did. But that is a claim about the limit. We do not work at that limit: at finite complexity and finite data, neither method can fit everything, and the extra structure in a causal probe might make it more resistant to fitting *spurious* targets, so causality could help in practice even if both methods fail in the limit. This report proposes measuring that directly—how well a method fits randomly relabelled data, which is a version of its empirical Rademacher complexity—in a form that applies unchanged to diagnostic probing and to DAS, and asks how this spurious-fitting capacity grows with complexity for each. We describe the measure and an experimental protocol on the hierarchical equality task, and report preliminary results: a linear probe recovers the task’s intermediate variables almost perfectly, while a linear DAS map does not recover them causally on the same model. We discuss why, and what is needed to turn this into a clean comparison.

1 Motivation

A probe is a supervised model trained to predict some property T from a network’s representations R ; if the probe succeeds, we are tempted to conclude that the representations *encode* the property ([Alain and Bengio, 2017](#); [Hewitt and Manning, 2019](#); [Belinkov, 2022](#)). [Pimentel et al. \(2020\)](#) recast this practice information-theoretically: probing estimates the mutual information $I(T; R)$, and the best estimate is obtained by the *best-performing* probe one can train, regardless of its complexity. The corollary is uncomfortable: under their [section 2](#) assumptions, a contextual representation contains exactly as much information about a word-level property as the underlying sentence does, so a powerful enough probe recovers the property for reasons that need have nothing to do with how—or whether—the network uses it.

Causal probing was meant to escape this critique by intervening rather than merely decoding. Distributed alignment search (DAS; [Geiger et al., 2024](#)) learns a (possibly non-linear) alignment map ϕ onto the hidden space, performs interchange interventions in the aligned basis, and measures

the *interchange-intervention accuracy* (IIA) with which the network reproduces a target algorithm’s counterfactual behaviour. Sutter et al. (2025) show that this escape is illusory: once the linearity constraint on ϕ is lifted, *any* algorithm can be aligned to *any* network with near-perfect IIA—even randomly initialised networks that cannot solve the task. They call this the **non-linear representation dilemma**: lifting the linearity constraint removes any principled way to trade off the alignment map’s complexity against its accuracy.

Read together, the two results are one failure seen twice: in each, an interpretability method reports a score—probe accuracy, or IIA—chosen as the best fit over a family of extraction functions, and in each, that score stops being informative once the family is expressive enough. The practitioner has no principled way to set the family’s complexity.

There is a pessimistic reading of this, and it is worth stating plainly. Sutter et al. (2025) show that in the limit of unconstrained alignment maps, causal abstraction is vacuous in the same way diagnostic probing is: the method succeeds regardless of what the network computes. At face value, this says that moving from diagnostic to causal probing buys us nothing—we are left where Pimentel et al. (2020) left us. But it is a claim about the limit, and it need not hold in practice. At the finite map complexities people actually use, the extra structure in a causal probe—it must act through the frozen network, and stay consistent across several interventions—might make it harder to fit spurious patterns than a diagnostic probe of comparable power. If so, causality does add something: not in the limit, but in the regime we work in. Whether it does is an empirical question. To answer it we need a measure of how readily each method fits spurious targets that applies to both—a single notion of capacity, in the spirit of Rademacher complexity (Bartlett and Mendelson, 2002) or VC dimension (Vapnik, 1998)—and a way to watch it grow with complexity.

2 Setup and notation

We adopt notation compatible with the two companion papers. A network induces a representation map $x \mapsto R$, with $R \in \mathbb{R}^d$. An **interpretability method** $\mathcal{M}$ is characterised by three ingredients: (i) a *hypothesis class* $\mathcal{H}_c$ indexed by a scalar *complexity knob* $c \in \mathbb{R}_+$; (ii) a per-example *fit* functional $\text{fit}(h; \cdot) \in [0, 1]$ measuring how well a hypothesis $h \in \mathcal{H}_c$ reproduces a target on one example; and (iii) a *selection rule* that, given a finite sample $\mathcal{D}_n$ of size n , returns $\hat{h} = \arg \max_{h \in \mathcal{H}_c} \hat{\text{fit}}(h; \mathcal{D}_n)$ and reports the achieved fit $s(\mathcal{M}) = \hat{\text{fit}}(\hat{h}; \mathcal{D}_n)$.

The two methods instantiate this template as follows.

Diagnostic probing. $\mathcal{H}_c$ is a family of probes $q_\theta(t \mid \mathbf{r})$ —e.g. multi-layer perceptrons whose depth or width grows with c , as in Pimentel et al. (2020). The targets are property labels t , the inputs are activations $\mathbf{r}$, and $\hat{\text{fit}}$ is accuracy (or, in the information-theoretic view, the negative cross-entropy that lower-bounds $I(T; R)$). The dataset is $\mathcal{D}_n = \{(\mathbf{r}_i, t_i)\}_{i=1}^n$.

Causal probing (DAS). $\mathcal{H}_c$ is a family of bijective alignment maps ϕ —e.g. reversible networks (Gomez et al., 2017) whose depth L_{rn} or hidden size d_{rn} grows with c , as in Sutter et al. (2025). The “label” of an example is the counterfactual output the *target algorithm* prescribes, the inputs are (base, source) pairs, and $\hat{\text{fit}}$ is the IIA: the rate at which the *frozen* network, intervened upon in the aligned basis, reproduces that output. The dataset is $\mathcal{D}_n = \{(\mathbf{x}_i^{\text{base}}, \mathbf{x}_i^{\text{src}}, \mathbf{y}_i^{\text{cf}})\}_{i=1}^n$.

The crucial structural difference—which we will argue is the heart of the matter—is that for DAS the hypothesis is not a classifier of $\mathbf{r}$. It is a *reparametrisation* ϕ of the hidden space, after which a *fixed* intervention and the *fixed* downstream network produce the prediction. The induced labelling function is $x \mapsto N(\text{intervene}_\phi(x))$, and the family is constrained to invertible ϕ .

3 A unified capacity measure

3.1 Spuriousness as fit to randomised targets

We want to measure how much of a method’s reported fit could be spurious—how well it fits targets that carry no real signal about the network. The direct way to do this is to relabel the data at random and check how well the method still fits it. The expected best fit a hypothesis class achieves on such random targets is, up to scaling, its empirical *Rademacher complexity* (Bartlett and Mendelson, 2002); we use it, in accuracy space, as our measure of spurious-fitting capacity.

Definition 1 (Spurious-fitting capacity). *Fix a representation sample of size n and let σ denote a target randomisation: a relabelling of the examples that destroys any genuine relationship between inputs and targets while preserving their marginals. The spurious-fitting capacity of a method $\mathcal{M}$ at complexity c is*

$$\mathfrak{R}_n(\mathcal{H}_c) := \mathbb{E}_{\sigma} \left[\sup_{h \in \mathcal{H}_c} \widehat{\text{fit}}(h; \sigma(\mathcal{D}_n)) \right] - s_{\text{chance}}, \quad (1)$$

i.e. the expected best fit the class achieves on randomised targets, centred at chance performance.

Equation 1 is, up to a scaling constant and the usual move from a margin loss to 0/1 accuracy, the empirical Rademacher complexity of $\mathcal{H}_c$ on the representation sample. Its appeal is threefold. First, it is *method-agnostic*: it specialises to diagnostic probing (random token labels) and to DAS (a random target algorithm, or equivalently permuted counterfactual outputs) without changing form. Second, it is exactly the quantity practitioners already estimate: it is the control-task accuracy of Hewitt and Liang (2019), and the random-network alignment of Sutter et al. (2025) is its causal analogue.¹ Third, it is *cheap*: a Monte-Carlo estimate requires only re-running the existing fitting pipeline on permuted targets.

3.2 The limit and the finite regime

1 is not what the two theorems are about, and the difference matters. The theorems are about a limit: at infinite complexity, both a probe and an alignment map can fit any target on any amount of data, random targets included (Pimentel et al., 2020; Sutter et al., 2025, Thm. 1). In that limit the two methods are equally uninformative and nothing tells them apart.

We never run at that limit. We fit at finite complexity on finite data, where neither method can fit everything. There, the two can differ: a linear probe and a linear rotation do not have the same ability to fit random relabellings, and that ability grows differently as we make each more complex. 1 measures exactly this ability, and its growth with complexity is what we study—not whether the methods break in the limit (they do), but how their capacity to fit spurious targets grows on the way there.

3.3 What “graceful” should mean, and the comparability problem

A natural temptation is to compare the two methods by plotting their capacity against a shared notion of “size”—say, parameter count—and asking whether one grows like $\mathcal{O}(\sqrt{P})$ where the other grows like $\mathcal{O}(P)$. We caution against taking the exponent itself as the contribution, for two reasons. First, generic norm- or parameter-based bounds yield the *same* $\mathcal{O}(\sqrt{P/n})$ form for any reasonable

¹We deliberately phrase 1 in accuracy space rather than as the textbook $\mathbb{E}_{\sigma}[\sup_h \frac{1}{n} \sum_i \sigma_i h(\mathbf{r}_i)]$ so that it is directly comparable across the two methods and directly estimable from the existing code paths, which already train on randomised labels.

class (Neyshabur et al., 2015), so an a priori difference in exponent is not expected; a measured difference would have to come from structure, not from counting. Second, “depth of an MLP probe” and “depth of a RevNet alignment map” are not the same unit, so a raw c -axis is not commensurable.

Both problems are solved by using *capacity itself* as the common axis. Concretely, we propose to plot, for each complexity level, the **signal** (fit on *correct* targets) against the **spurious capacity** (fit on *randomised* targets, 1). Each method traces a curve in this *signal–spurious plane* as c varies. A method **fails gracefully** if its curve hugs the top-left—high signal recovered per unit of spurious capacity spent—and **fails sharply** if signal and spurious capacity rise together. This operationalises “one method fails more gracefully than the other” without ever committing to incommensurable units, and it directly generalises the selectivity of Hewitt and Liang (2019) (which is the single-point gap signal – spurious) into a *scaling curve*.

4 A mechanism for divergence

If the two methods do diverge in the signal–spurious plane, the explanation must lie in structural constraints that one method imposes and the other does not. We identify three constraints that are present in DAS but absent in diagnostic probing, each of which should *throttle* spurious capacity below what the raw parameter count of ϕ would suggest:

1. **Invertibility.** The alignment map ϕ is constrained to be a bijection. A bijection cannot collapse or discard directions of the hidden space to manufacture a fit; it can only rotate and warp them. This is a genuine capacity restriction with no analogue in an unconstrained probe.
2. **The frozen network.** The prediction is produced by the *fixed* downstream network applied to an intervened, ϕ -aligned hidden state. The map cannot directly emit a label; it can only reroute information that the frozen weights will then process. Spurious fits must survive this fixed computation.
3. **Multi-intervention consistency.** A single ϕ must simultaneously satisfy the counterfactual constraints of *several* intervention types (in the hierarchical equality task, the two sub-equalities and their conjunction). These are many coupled constraints, not one label, and jointly satisfying them on randomised targets is harder than fitting a single randomised label.

Our working hypothesis is that these constraints make $\mathfrak{R}_n^{\text{DAS}}(c)$ grow *more slowly* than $\mathfrak{R}_n^{\text{DP}}(c)$ —i.e. that DAS, despite the alarming $c \rightarrow \infty$ result of Sutter et al. (2025), may be the more graceful failure at the finite capacities used in practice. We stress that this is a hypothesis to be tested, not a claim; the constraints above could equally be swamped by the sheer flexibility a non-linear ϕ buys, in which case DAS would fail *more sharply*. Either outcome is informative.

5 Experimental protocol

We instantiate the study on the **hierarchical equality task**, the toy setting of Sutter et al. (2025), for which an implementation already exists in this repository (a 3-layer MLP trained to decide whether two pairs of vectors are “both equal or both unequal”, together with linear-probe and linear-DAS pipelines and *randomised-label* controls). The task is ideal because the ground-truth intermediate variables (WX, YZ) are known, so “signal” is unambiguous.

Complexity axes. We add a complexity knob to each method. For *diagnostic probing*, we replace the single logistic-regression probe with a family of MLP probes of increasing depth and width ($c \in \{\text{linear}, 1\text{-hidden-layer}, 2\text{-hidden-layer}, \dots\}$), as in Pimentel et al. (2020). For *DAS*, we replace the orthogonal (rotation) alignment map with a family of RevNet maps of increasing depth L_{rn} and hidden size d_{rn} , as in Sutter et al. (2025). The data budget n is held *fixed* across the sweep so that we isolate the effect of c (in contrast to the existing sweeps, which vary n).

Measurements. At each complexity level and for several seeds we record (i) the *signal*: fit on correct targets (probe accuracy / IIA on the trained network); and (ii) the *spurious capacity* $\hat{\mathfrak{R}}_n(\mathcal{H}_c)$ of 1, estimated by re-fitting on randomised targets (random token labels for probing; a randomly relabelled / random target algorithm for DAS). We additionally repeat the DAS measurement on a *randomly initialised* network, which makes the spurious axis vivid: any IIA there is, by construction, spurious.

Analysis. We (1) plot signal and spurious capacity against c for each method; (2) plot the two methods together in the *signal-spurious plane* of section 3; and (3) fit a simple growth model $\hat{\mathfrak{R}}_n(c) \approx a c^b$ to read off, and compare, the empirical growth exponents b with confidence intervals. The deliverable is a statement of the form “per unit of spurious capacity, method X recovers more genuine signal than method Y on this task,” with the supporting curves.

What would falsify the framing. If signal and spurious capacity rise in lockstep for *both* methods (curves collapsing onto the diagonal of the signal-spurious plane), then capacity carries no method-distinguishing information and the unification, while tidy, would be practically empty. We regard making this falsifiable as a feature.

6 Preliminary results

We report initial measurements on the hierarchical equality task. They are preliminary: we have a clean diagnostic-probe baseline and a working DAS pipeline, but the full complexity sweep (deeper probes and RevNet alignment maps) is not yet run.

The intermediate variables are linearly decodable. A linear probe on the trained MLP’s first-layer activations recovers WX and YZ with about 0.98 and 0.96 accuracy. Decodability decreases with depth (to about 0.75 by the third layer), while the output O becomes near-perfectly decodable only in the deeper layers ($0.70 \rightarrow 0.997$). So at the layer where the two variables are most linearly present, a probe finds them almost perfectly.

Linear DAS does not recover them causally. Training a linear (rotation) alignment map at that same layer to perform the WX interchange yields interchange-intervention accuracy at roughly chance ($\sim 0.52\text{--}0.56$), even when we train on the single, label-balanced WX intervention in isolation. This is not an optimisation failure of the obvious kind: the rotation receives a non-trivial gradient and moves steadily away from its initialisation throughout training; the accuracy simply does not improve. So on this model a feature that is almost perfectly *decodable* by a linear probe is not *interveneable* by a linear alignment map.

We are cautious about over-reading this DAS number. It is below the interchange accuracy that linear DAS reaches on this task in Sutter et al. (2025), which points to a factor specific to our setup rather than a property of linear alignment in general. The most likely culprit is that our trained

MLP is badly calibrated: it produces very large logits (we observed class-logit gaps of order 50, with cross-entropy of order 10 under a random alignment), which makes the cross-entropy landscape the rotation is optimised on sharp and hard to descend. Calibrating the target model is the first thing to fix before reading anything into the DAS numbers.

7 Implementation notes

Two methodological points cost us the most time and are worth recording, as they affect anyone running DAS on this kind of data.

Judge causal probes by held-out, order-independent evaluation. The obvious training-time metric—a running average of interchange accuracy—is misleading here. The counterfactual data is generated in contiguous blocks that each share one intervention type, and for the joint (both-subspace) intervention each block turns out to be label-homogeneous. A running average over such blocks drifts with the order the blocks happen to arrive in, which looks exactly like a model that “learns then collapses” even when the alignment map is frozen. Every number above comes from a single order-independent pass over a held-out set, broken down by intervention type.

The joint intervention is not informative as generated. Because each joint-intervention block is a single class, it provides no signal for learning the equality, and its per-block accuracy merely tracks the block’s label; we therefore do not report it. Balancing labels within each block (or mixing blocks within a gradient step) is needed before the joint intervention can be trained or evaluated meaningfully. Two smaller pitfalls: a “fast” intervention path we used initially keeps only the first of several source locations, silently dropping the second source in the joint intervention; and, as noted above, the target model’s miscalibration is a likely cause of the poor DAS optimisation and has not yet been isolated.

8 Relation to prior work, and a caveat on consistency

The control-task literature is the obvious antecedent. [Hewitt and Liang \(2019\)](#) introduce *selectivity*—task accuracy minus control-task accuracy—as a criterion for choosing probes, and [Voita and Titov \(2020\)](#) propose minimum description length as a complexity-aware alternative. Our spurious-fitting capacity is, at a single complexity level, exactly a control-task score; the novelty here is (i) reading it as a Rademacher complexity, (ii) lifting it from diagnostic to *causal* probing, and (iii) studying its *scaling* rather than its value at one point.

We must, however, confront a tension with [Pimentel et al. \(2020\)](#), which argues *against* selectivity as a probe-*selection* criterion: penalising a complex probe for its control-task accuracy artificially discards capacity that yields a tighter information estimate. We do not retract that argument; we repurpose the quantity. Selectivity is a poor *objective* to optimise—one should still pick the best-performing probe for estimation—but it is an excellent *descriptive diagnostic* of how much a method *could* fool itself, and its growth rate is a property of the method, not a knob to tune. Measuring fragility is not the same as selecting against it.

Finally, a caveat on what [1](#) measures. It is about fitting random targets at finite data and finite complexity. This is related to, but not the same as, what the two theorems prove—that at infinite complexity any target can be fit. A method could be reluctant to fit random targets at the complexities we actually test and still be able to fit anything in the limit. Our measure speaks to the former, the regime we use in practice, not the latter.

9 Conclusion

We have argued that the diagnostic critique of Pimentel et al. (2020) and the causal critique of Sutter et al. (2025) are two instances of the same thing: a method whose reported fit grows with its capacity to fit anything, spurious targets included. We proposed measuring that capacity directly—how well a method fits randomly relabelled data—in a form that applies to both diagnostic probing and DAS, and studying how it grows with complexity. Our preliminary results on the hierarchical equality task show a sharp contrast on the causal side: the task’s intermediate variables are almost perfectly linearly decodable, yet a linear alignment map does not recover them causally on the same model. We suspect this last result is partly an artefact of a miscalibrated target model rather than a property of linear alignment, and resolving that is the first step before the full complexity sweep can be run. The payoff, if the sweep bears it out, is a concrete sense in which causal probing is the safer tool at finite complexity—even though, in the limit, it is no better than the diagnostic probing it was meant to improve on.

References

- Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. In *International Conference on Learning Representations (Workshop)*, 2017.
- Peter L. Bartlett and Shahar Mendelson. Rademacher and Gaussian complexities: Risk bounds and structural results. *Journal of Machine Learning Research*, 3:463–482, 2002.
- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.
- Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah D. Goodman. Finding alignments between interpretable causal variables and distributed neural representations. In *Causal Learning and Reasoning*, 2024.
- Aidan N. Gomez, Mengye Ren, Raquel Urtasun, and Roger B. Grosse. The reversible residual network: Backpropagation without storing activations. *Advances in Neural Information Processing Systems*, 2017.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 2733–2743, 2019.
- John Hewitt and Christopher D. Manning. A structural probe for finding syntax in word representations. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 4129–4138, 2019.
- Behnam Neyshabur, Ryota Tomioka, and Nathan Srebro. Norm-based capacity control in neural networks. *Conference on Learning Theory*, 2015.
- Tiago Pimentel, Josef Valvoda, Rowan Hall Maudslay, Ran Zmigrod, Adina Williams, and Ryan Cotterell. Information-theoretic probing for linguistic structure. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4609–4622, 2020.
- Denis Sutter, Julian Minder, Thomas Hofmann, and Tiago Pimentel. The non-linear representation dilemma: Is causal abstraction enough for mechanistic interpretability? In *Advances in Neural Information Processing Systems*, 2025.
- Vladimir N. Vapnik. *Statistical Learning Theory*. Wiley, 1998.
- Elena Voita and Ivan Titov. Information-theoretic probing with minimum description length. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 183–196, 2020.